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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025**25VDE122 – SYSTEM VERILOG FOR VERIFICATION AND TESTING**

(For VDE)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

		Marks	BT*	CO*	PO*
1.	a) Discuss the 2-state vs 4-state integer data types with an example.	10	2	1	3
	b) Explain the behavior of a System Verilog dynamic array with an appropriate code.	10	3	1	3
2.	a) Illustrate the System Verilog associative array with an appropriate code.	10	3	2	3
	b) Explain parallel blocks in System Verilog. What are the types of parallel blocks available? Write suitable examples for each type.	10	2	2	3
3.	a) Illustrate the different types of bins and the types of transitions in bins in System Verilog w.r.t. Functional Coverage.	10	2	1	3
	b) What are the jumping and looping statements in system Verilog. Explain each one of them with an example.	10	2	2	3

Module – II

4.	a) Compare uvm_object and uvm_component.	10	2	3	3
	b) Justify the need for UVM with an example.	10	2	3	3
5.	a) With a neat diagram explain the UVM components and its purpose.	10	2	3	3
	b) Demonstrate how randomization control can be applied to generate constrained test sequences for verifying a communication protocol.	10	3	4	3
6.	a) Compare create and new(). With examples.	10	2	4	3
	b) Illustrate UVM agent with an example.	10	2	4	3

Module – III

7.	a) Explain the UVM Testbench Connection example with a code.	10	3	5	3
	b) Illustrate the use of UVM messaging macros such as uvm_info, uvm_warning, and uvm_error. How do they assist in debugging the verification environment? Give a suitable example.	10	2	5	3
8.	a) Explain the example of functional coverage of a simple object.	10	2	5	3
	b) Illustrate the UVM Messaging example with a code.	10	3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
